Centre for Cognitive Ageing and Cognitive Epidemiology

Is feature binding in visual working memory age invariant?

Stephen Rhodes[1], Mario A. Parra[1], Nelson Cowan[1,2], & Robert H. Logie[1]

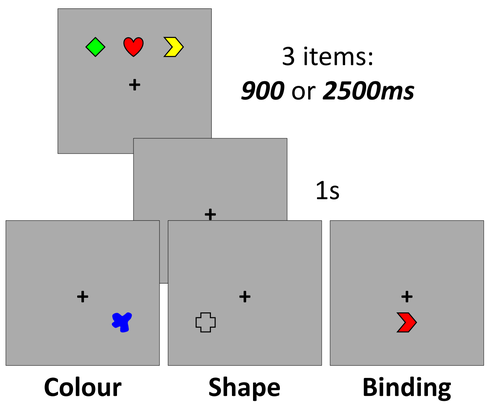
[1]University of Edinburgh, [2]University of Missouri-Columbia

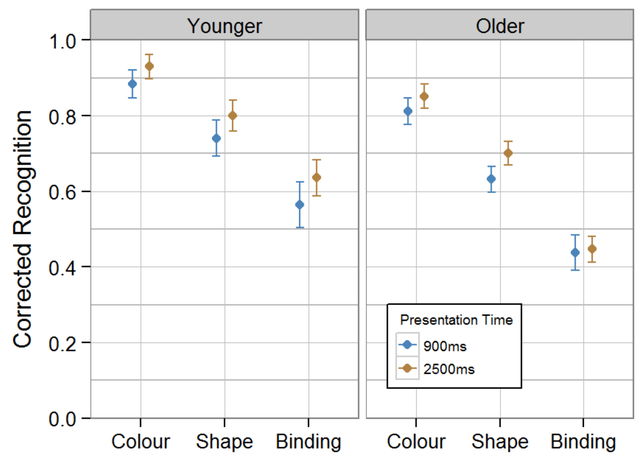
Introduction

- Visual working memory (VWM) declines substantially across the adult life-span [1].
- Suggested that age may impair ability to bind different object features (e.g. colour, shape, location) into integrated representation in VWM [2, 3].
- Little previous evidence to suggest VWM for colour-shape combinations more age-sensitive than memory for these features individually [4, 2]. Particularly notable given the pronounced feature binding deficit seen in Alzheimer's disease [5, 6].
- Present experiments aimed to test whether there are conditions under which feature binding in VWM is not entirely age invariant [7, 3].
- Change detection task for individual features and feature combinations. 50% of trials were change trials (pictured below), 50% no-change.
- In order to assess the evidence *against* age-group interactions we report Bayes factors (BF)† for inference [8].

Experiment 1 – Presentation Time

- Brown & Brockmole [7]: age-related binding deficit at 1500ms, not 900ms presentation time (PT). PT was not their main focus.
- We directly assessed effect of PT on younger and older adults' change detection.
- **Method:** 24 young (*M* = 21.37, *SD* = 2.10) and 24 older (73.17, 3.69). Memory array presented for 900 or 2500ms.





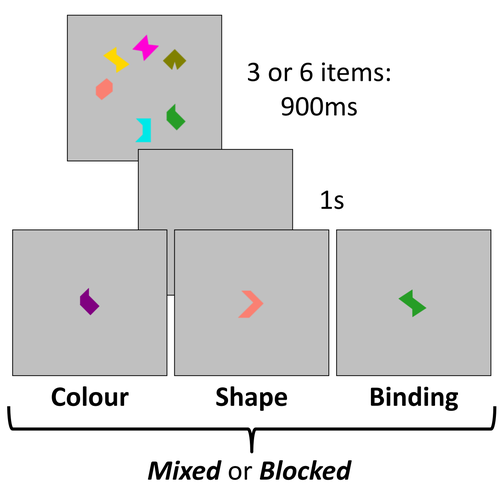
Age × condition interaction:
BF‡ = 1.92 *against* the interaction (data approximately 2 times more likely under null).
3 way interaction:
BF = 6.02 against the interaction

Increasing presentation time does not differentially affect younger and older adults' performance in different conditions.

Experiments 2A and 2B – Mixed vs. Blocked changes

2A – colour and shape

- Cowan et al. [3]: small age-related colour-location binding deficit when conditions blocked.
- Age effect larger for mixed conditions. Older adults tended to miss binding changes.
- Possibly increased reliance on familiarity.
- Also true for colour-shape binding?
- **Method:** 36 young (21.19, 2.9) and 36 older (69.03, 3.02). Half see mixed, half blocked.

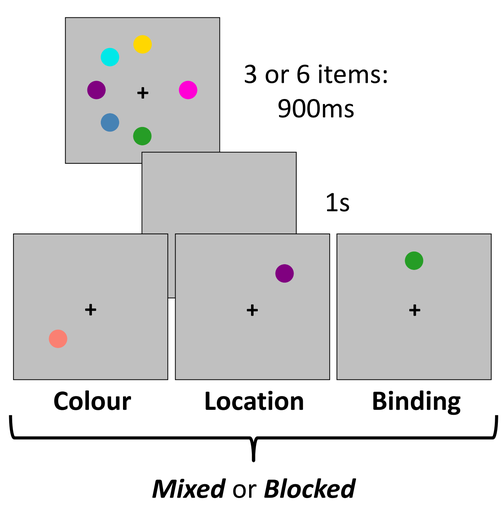


Age × Condition:
BF = 17.26 against interaction
Age × Condition × Set size:
BF = 1.54 against
Age × Condition × Block type:
BF = 11.08 against
4 way interaction:
BF = 23.52 against

Strong evidence against a differential effect of age on VWM performance in different conditions.

2B – colour and location

- Something special about colour-location binding that leads to an age effect?
- Cowan et al. [3]: Colour and binding changes only.
- Possible bias towards one feature over the other at the expense of binding.
- Is colour-location binding deficit present with inclusion of all 3 types of change trial?
- **Method:** 20 young (20.15, 2.41) and 20 older adults (71.35, 4.77). Half see mixed, half see blocked.



Age × Condition:
BF = 10.83 against interaction
Age × Condition × Set size:
BF = 2.81 against
Age × Condition × Block type:
BF = 7.36 against
4 way interaction:
BF = 11.75 against

Preliminary data provides substantial-strong evidence against a differential effect of age on VWM performance in different conditions.

- Across 3 experiments, no evidence for a differential effect of age on VWM for individual features and bindings. Rather, Bayes factors consistently provide evidence against age-group by condition interactions.
- Recently suggested there may be a small, unreliable age-related feature binding deficit [10]. Clearer evidence will come from amalgamation of findings via meta-analytical techniques. But present data inconsistent with age-related feature binding deficit in VWM.

Notes:† Bayes factors represent the weight of evidence for one hypothesis (e.g. an interaction) versus another (e.g. main effects only – null). If BF = Pr(data | H_1) / Pr(data | H_0), then BF > 1 = support for H_1, whereas BF < 1 = support for H_0. As a general rule BFs ≥ 3 are considered 'substantial evidence' [11]. ‡ Bayes factors were calculated using the `BayesFactor` package (http://CRAN.R-project.org/package=BayesFactor) in `R` (http://www.R-project.org/).

References: **[1]** Johnson, W. et al. (2010). *Intelligence*, *38* , 513. **[2]** Brockmole, J. R. et al. (2008). *Psychon. Bull. & Rev*, *15*, 543. **[3]** Cowan, N. et al. (2006). *Dev. Psychol.*, *42*, 1089. **[4]** Parra, M. A. et al. (2009). *Neurosci. Lett.*, *449* , 1. **[5]** Parra, M. A. et al. (2009). *Brain*, *132*, 1057. **[6]** Della Sala, S. et al. (2012). *Neuropsychologia*, *50* , 833. **[7]** Brown, L. A., & Brockmole, J. R. (2010). *Q J Exp. Psych.*, *63*, 2067. **[8]** Rouder, J. N. et al. (2012). *J Math. Psychol.*, *56*, 356. **[9]** Allen, R. J. et al. (2006). *J Exp. Psychol: Gen.*, *135*, 298. **[10]** Allen, R. J. et al. (2013). In H. St. Clair-Thompson (Ed.), *Working memory* (pp. 83–96). Nova Science Publishers. **[11]** Jeffreys, H. (1961). *Theory of probability (3rd ed.)*. Oxford University Press.

Acknowledgements: This work was undertaken by the University of Edinburgh Centre for Cognitive Ageing and Cognitive Epidemiology, part of the cross council Lifelong Health and Wellbeing initiative. Funding from the BBSRC, EPSRC, ESRC and MRC is gratefully acknowledged (G0700704).